APPENDIX

A. Summary of Key Notations

To facilitate understanding, key notations used in this paper are listed in Table IV.

Table IV List of some key notations.

Notation	Description
VirEqPhy	The proposed method for geometric dynamic equivalence between virtual and physical spaces which mainly includes multidimensional feature extraction module **MI**, voxel representation learning module **MV**, model embedding retrieval module **MR** and model cross-domain alignment module **MA**.
$\boldsymbol{Im}$, **DS**	RGB image that describes the real scene observed by the agent and digital space corresponding to the real scene.
$\boldsymbol{E}, \boldsymbol{DB}$	Set of objects $\{obj_1, \dots, obj_{\lvert E\rvert}\}$ in the real scene and CAD model library $\{cad_1, \dots, cad_{\lvert \mathrm{DB}\rvert}\}$ of objects.
$\boldsymbol{F}_{\mathrm{I}}, \boldsymbol{F}_{\mathrm{V}}$	Image representation of $\boldsymbol{Im}$ and voxel representation of objects. The former includes image depth $\boldsymbol{D}$, object class c, bounding box $\boldsymbol{b}$, and mask $\boldsymbol{m}$; The latter contains 3D voxel grid $\boldsymbol{v}_{obj}$ and voxel coordinates $\boldsymbol{n}_{obj}$ of objects.
$\boldsymbol{s}, \boldsymbol{t}, \boldsymbol{r}$	Scale, position, and rotation of objects.
$\boldsymbol{SM}_{\mathrm{gt}}^{\mathrm{asg}}$	Normalized similarity matrix of all objects and CAD models in weakly supervised datasets based on appearance similarity $S_{\mathrm{gt}}^{\mathrm{app}}$, semantic similarity $S_{\mathrm{gt}}^{\mathrm{sem}}$, and geometric similarity $S_{\mathrm{gt}}^{\mathrm{geo}}$. Each similarity is calculated separately through appearance feature $\boldsymbol{f}^{\mathrm{app}}$, semantic feature $\boldsymbol{f}^{\mathrm{sem}}$, and voxel feature $\boldsymbol{f}^{\mathrm{vox}}$.
$\boldsymbol{SM}, \boldsymbol{I}$	Normalized similarity matrix and its indicator matrix between objects and CAD models.

B. Additional Results for *VirEqPhy*

Table V Retrieval Accuracy (%) of various methods. **AP50** and **AP75** denote retrieval AP at the corresponding evaluation thresholds, and **APm** denotes the mean AP over thresholds from 0.50 to 0.95. The blue background indicates the top three values.

Method	bed	sofa	chair	bin	cabinet	display	table	bk shlf	Apm (mean)	AP50 (mean)	AP75 (mean)
Total3D	1.9	4.3	1.5	0.8	0.1	0.0	0.7	2.1	1.4	6.3	0.2
Mask2CAD	14.2	13.0	13.2	7.5	7.8	5.9	2.9	3.1	8.4	22.5	4.9
Patch2CAD	18.8	12.4	17.6	7.5	8.6	10.8	3.3	3.3	10.3	20.6	6.6
ROCA	15.6	11.7	17.1	6.5	8.8	9.2	5.4	3.5	10.0	19.4	5.5
FastCAD	15.5	12.0	17.5	6.4	8.2	8.5	4.8	3.7	9.8	20.5	5.5
DiffCAD	20.2	11.6	20.1	6.0	10.3	10.7	5.5	5.3	11.6	23.9	6.6
Ours	19.1	13.0	21.9	7.2	9.4	10.5	5.6	3.8	11.3	24.2	6.7

Table VI Alignment Accuracy (%) of various methods. Class and Instance represent the average accuracy over the class and instance, respectively. The blue background indicates the top three values.

Method	bed	sofa	chair	bin	cabinet	display	table	bk shlf	bathtub	class	instance
Mask2CAD	2.9	5.3	30.9	25.9	5.4	17.3	7.1	3.8	8.3	11.9	17.9
ROCA	10.0	15.9	41.0	29.3	15.8	30.4	14.6	14.2	22.5	21.5	27.4
FastCAD	17.7	19.1	44.9	30.6	16.1	24.7	17.8	16.5	20.4	23.2	27.6
DiffCAD	19.5	23.3	45.7	28.9	18.4	25.2	23.2	19.5	20.7	23.8	28.8
InCAD	18.6	38.1	49.3	22.8	9.2	24.1	16.5	12.7	16.7	23.1	30.1
Ours	16.3	29.8	48.7	27.8	18.6	31.2	24.6	19.5	23.3	24.4	29.6

Tables V and VI report the Retrieval Accuracy and Alignment Accuracy of the various methods, respectively. From Table V, overall, our method achieves the best high-threshold retrieval and the second-best mean retrieval, reflecting that the voxel-based joint embedding space favors tighter shape-pose matches at stricter IoU thresholds. **DiffCAD** is highly competitive under weak supervision and tends to lead on categories with regular, stable geometry (e.g., box-like furniture), where probabilistic multi-hypothesis matching readily captures near-duplicate shapes. **Patch2CAD**, with its patchwise feature embedding, attains solid retrieval accuracy, particularly on smaller or edge-dominated objects such as *bin* and *display*. **ROCA** benefits from a 3D coordinate joint embedding that improves robustness over classic two-stage methods (e.g., **Mask2CAD**), but without explicit pose refinement in the retrieval objective, its gains remain limited. As shown in Table VI, our method achieves the best class-level and second-best instance-level alignment, indicating stronger global spatial consistency under the joint position/rotation/scale criterion. It is particularly effective on geometry-driven categories (*cabinet*, *display*, *table*, *book-shelf*), owing to voxelized geometry and a two-stage rotation scheme that reduces pose ambiguity and couples retrieval with pose for tighter alignment. Notably, **InCAD** performs strongly on shape-rich categories (e.g., *sofa*, *chair*) via geometry-aware learning but underperforms on box-like categories with uniform shape (e.g., *cabinet*). **DiffCAD** remains broadly stable across categories — especially *bed* and *book-shelf* — owing to its diffusion-based probabilistic modeling, which improves global alignment consistency.

Fig. 12 further visualizes inference results for occluded objects. Although some detail loss occurs, our method's predicted voxel representations retain richer visual information than 2D features, enabling retrieval of the most similar CAD models.

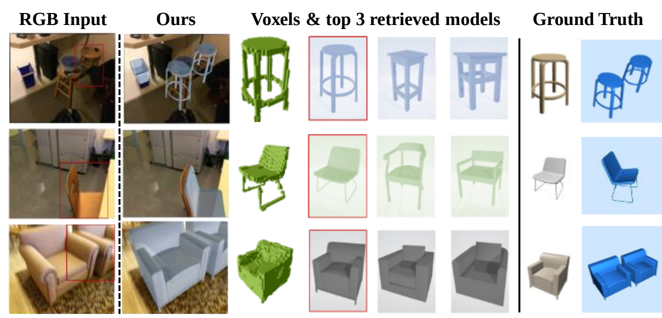

Fig. 12. Qualitative results in scenarios with partially occluded objects.

C. Qualitative Results for **W-MR**

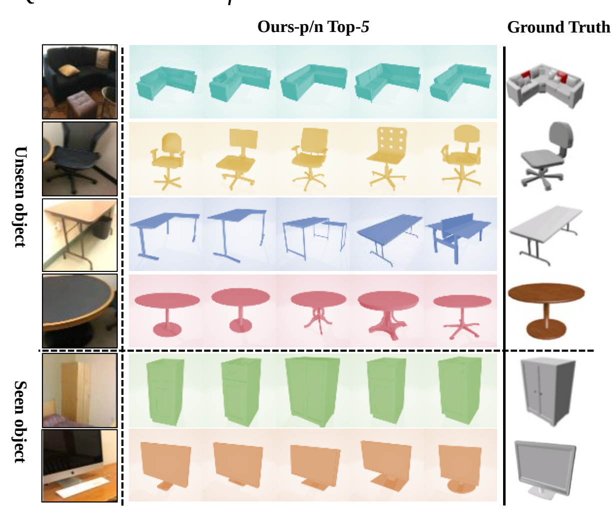

Fig. 13. Qualitative results on some seen and unseen objects.

Fig. 13 further illustrates **W-MR**'s performance on seen and unseen objects of different categories across multiple viewpoints (K=5). For seen categories, retrieved CAD models closely match ground-truth geometry and scale. For unseen objects, top-5 retrieved models accurately reflect class characteristics. Although minor geometric mismatches occasionally occur (e.g., chair backrest/armrests shapes in the second instance), top-5 retrieved models still capture overall geometric and semantic characteristics of the target object. Moreover, even under occlusion (e.g., the first instance), the retrieval result remains reasonable, confirming the compatibility of voxel-based feature embedding with **W-MR**.

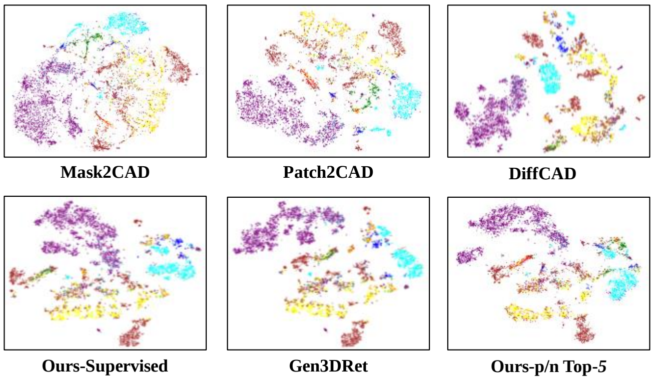

Fig. 14. *t-SNE* visualization of embedding space of CAD models. The same color represents the same class of CAD models.

Fig. 14 provides a *t-SNE* visualization of the embedding space after training, illustrating how different methods group CAD models. From the figure, it can be seen that **W-MR** forms tighter intra-class clusters and clearer inter-class separations than the baselines, indicating a more discriminative retrieval space. This enables the retrieved models to exhibit more precise separation between intra-class and inter-class instances. This qualitative result aligns with Table II, clearly indicating that the **W-MR** can enhance the model retrieval ability.

Fig. 15 visualizes the most similar CAD models (intra-class) and least similar CAD models (inter-class) retrieved by **Our-p/n Top-*1*** for unseen objects during training. The most similar CAD models exhibit nearly identical geometric and semantic characteristics to target objects, while the least similar models (used as negative examples) belong to distinct categories or have dissimilar shapes. This confirms that our method with positive/negative samples not only identifies optimal matches but also learns dissimilar boundaries.

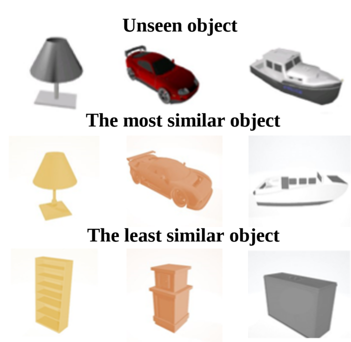

Fig. 15. Most similar and least similar CAD models of some unseen objects.

A dynamic visualization in Fig. 16 shows that, as the camera moves, our method consistently retrieves the correct CAD models and refines 9-DoF poses in real time. In conclusion, this CAD model retrieval enhancement module reduces manual annotation cost while improving ***VirEqPhy***'s robustness and accuracy for unseen object. This generalization ability is essential for expanding digital twin systems, enabling adaptation to dynamic environments.

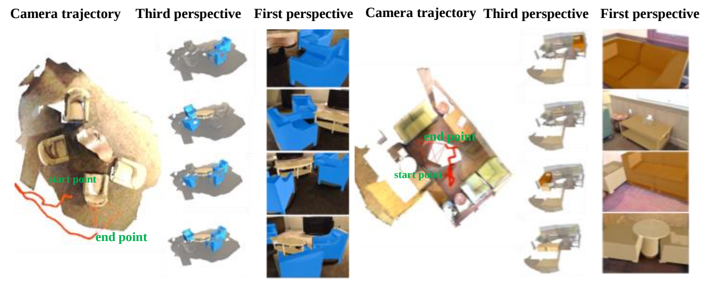

Fig. 16. Dynamic visualization results.

D. *Performance of Cross-Domain Transfer*

To further evaluate the cross-domain transferability of the proposed ***VirEqPhy***, we conduct an additional experiment on the high-fidelity simulation platform AI Habitat. It is worth noting that cross-domain evaluation in AI Habitat offers several advantages: it enables controllable, reproducible shifts (lighting, layout, object mix), supplies precise annotations (RGB/depth/semantic and 9-DoF), and provides low-cost scalability to add new CADs for efficient multi-scene generalization tests.

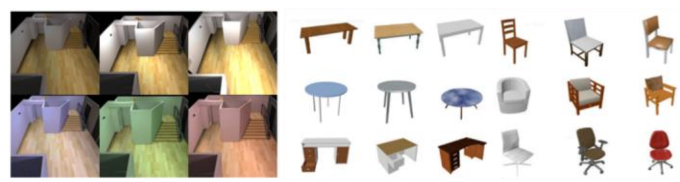

Fig 17. Different types of light sources and examples of the new CAD models.

1) Experimental Settings We create 60 virtual indoor scenes by introducing various light sources, additional CAD models (Fig. 17), and randomly distributing small objects (e.g., bowls, books) to increase scene complexity. Each scene is captured by a pinhole camera mounted on the agent, recording RGB, depth, and semantic images under different lighting conditions, resulting in 2,400 data samples. ***VirEqPhy*** is trained on the Scan2CAD-based dataset (i.e., ScanNet-ShapeNet alignment) and then evaluated on synthetic scenes created in AI Habitat.

Table VII Retrieval-Aware Alignment Accuracy (%) on different datasets.

Method	**Scan2CAD-based dataset**		**Habitat-based synthetic dataset**	
	class	**instance**	**class**	**instance**
Mask2CAD	**9.5**	**13.8**	**18.6**	**20.0**
ROCA	**17.6**	**21.7**	**23.2**	**25.1**
DiffCAD	**19.8**	**24.6**	**24.8**	**27.0**
FastCAD	**18.5**	**21.4**	**23.2**	**25.0**
InCAD	**19.4**	**23.0**	**24.5**	**26.4**
Ours	**20.6**	**24.5**	**25.2**	**27.8**

2) Results The results in Table VII show that all methods perform better on the synthetic dataset, primarily due to more balanced object distributions and controlled lighting and layout. Our method achieves state-of-the-art performance on both class-level and instance-level metrics, indicating robust generalization under shifts in lighting, layout, and object composition. We attribute this improvement to preserving fine-grained geometric cues and tightly coupling retrieval with alignment in a unified framework.

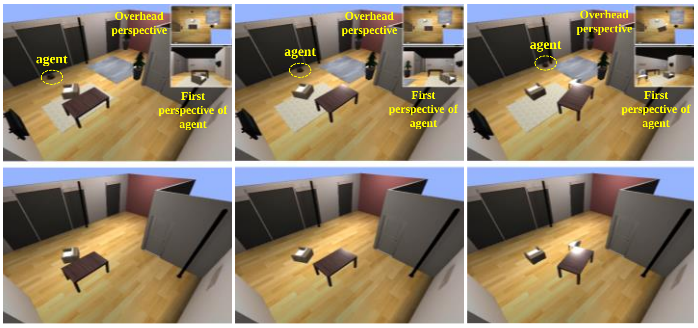

Fig 18. Top: Scene changes in Habitat, Bottom: Scene updating based on the proposed method.

Fig. 18 illustrates the scene dynamics and the corresponding digital-twin updates during continuous indoor navigation. Despite viewpoint changes, occlusion by tables and sofas, and the introduction of a new object (a chair), the system consistently retrieves the correct CAD models and maintains accurate 9-DoF alignment. These results further demonstrate effective cross-domain transfer beyond the original data domain.

E. Case Study on Multi-Agent Restaurant Service

We conditionally deploy the proposed ***VirEqPhy*** in a restaurant-service scenario, where multiple heterogeneous agents autonomously navigate and collaboratively deliver, place, and retrieve tableware and food items of varying shapes and sizes to enhance the dining experience. For these non-standard objects, we generate CAD models via 3D reconstruction and augment the retrieval database accordingly. Experimental results indicate that ***VirEqPhy*** tracks geometric changes in the physical space in real time with satisfactory accuracy in this setting. Due to space limitations, additional details are provided in the supplementary material and at https://forbiddenname.github.io/VirEqPhyDT.